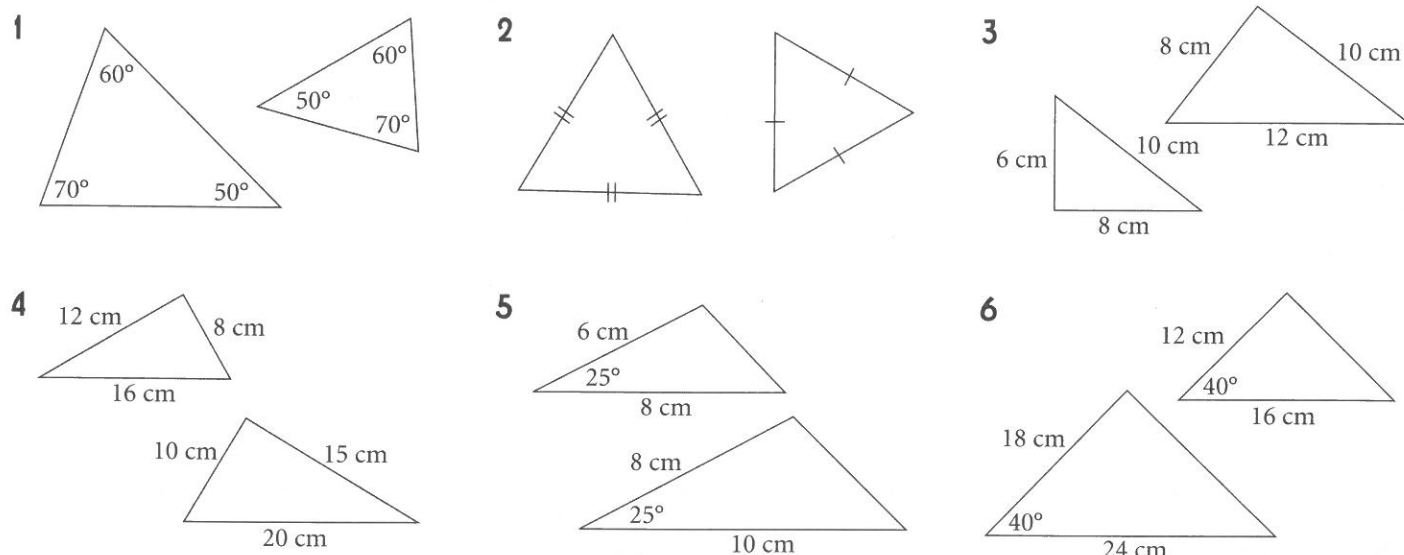
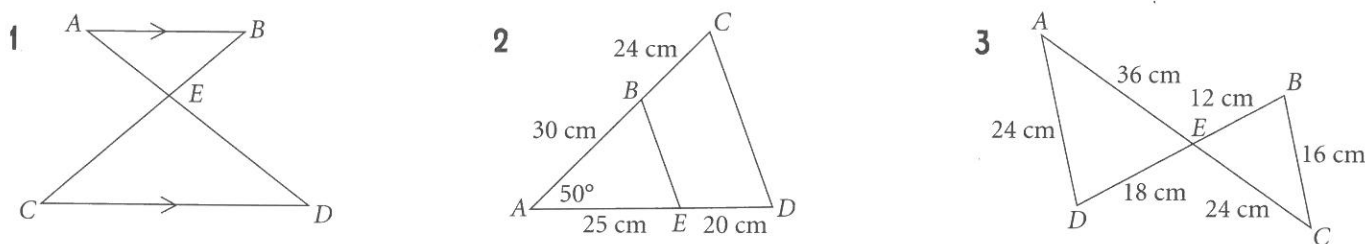


A Are the following pairs of triangles similar?

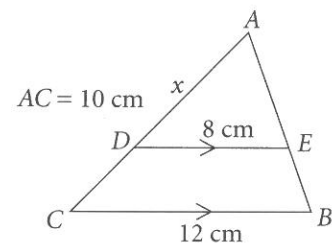


B Name two similar triangles in each of the following diagrams. Briefly explain why they are similar.



Example:

In the triangle ABC , $DE \parallel CB$ and $AC = 10$ cm. Prove that $\triangle ACB$ is similar to $\triangle ADE$ and find the value of x .



Solution:

In the $\triangle s$ ACB and ADE , $\angle BAC$ is common,
 $\angle ACB = \angle ADE$ (equal corresponding angles, $CB \parallel DE$)
 and $\angle ABC = \angle AED$.

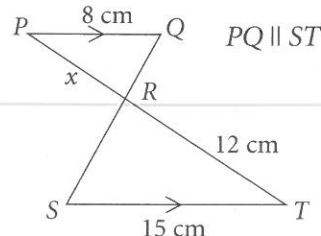
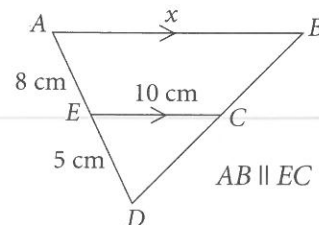
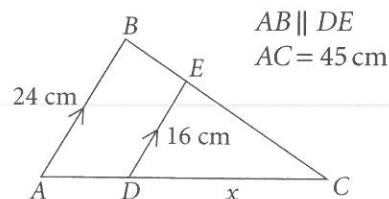
$\therefore \triangle ACB$ is similar to $\triangle ADE$ (equiangular)

$\therefore \frac{AD}{AC} = \frac{DE}{CB}$ (ratio of corresponding sides)

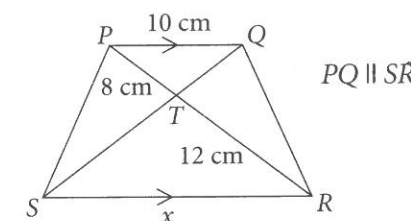
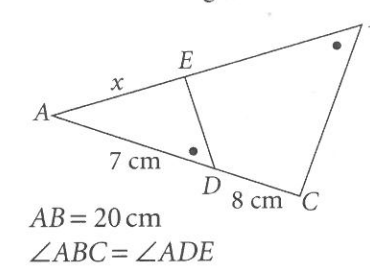
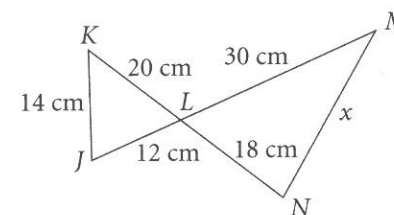
$$\therefore \frac{x}{10} = \frac{8}{12}$$

$$\therefore x = 10 \times \frac{8}{12} \\ = 6.7 \text{ cm (to 1 decimal place)}$$

- 4 a Prove that $\triangle ABC$ is similar to $\triangle DEC$.
 b Find the length of DC .
- 5 a Prove that $\triangle ABD$ is similar to $\triangle ECD$.
 b Find the length of AB .
- 6 a Prove that $\triangle PQR$ is similar to $\triangle TSR$.
 b Find the length of PR .



- 7 a Prove that $\triangle JKL$ is similar to $\triangle NML$.
 b Find the length of MN .
- 8 a Prove that $\triangle ABC$ is similar to $\triangle ADE$.
 b Find the length of AE .
- 9 a Prove that $\triangle PQT$ is similar to $\triangle RST$.
 b Find the length of SR .



C

- 1 D is the midpoint of AB and $ED \parallel CB$. Prove that:
 a $\triangle ABC$ is similar to $\triangle ADE$
 b $AE = EC$
- 2 Straight lines AB and CD intersect at the point E . Prove that:
 a $\triangle ACE$ is similar to $\triangle BDE$
 b $AC \parallel DB$
- 3 Prove that:
 a $\triangle ABC$ is similar to $\triangle DAC$
 b $\angle BCA = \angle DCA$
- 4 Straight lines PQ and RS intersect at the point T .
 a Prove that $\triangle PRT$ is similar to $\triangle SQT$.
 b Is $PR \parallel QS$?
- 5 X and Y are the midpoints of AB and BC respectively. Prove that:
 a $\triangle ABC$ is similar to $\triangle XBY$
 b $AC \parallel XY$
- 6 $ABCD$ is a rectangle. AE is drawn perpendicular to the diagonal DB . Prove that:
 a $\triangle ADE$ is similar to $\triangle DBC$
 b $AD \times DC = DB \times AE$
- 7 PQR and PSQ are right-angled triangles. Prove that:
 a $\triangle PQR$ is similar to $\triangle PSQ$
 b $PQ^2 = PS \times PR$
- 8 O is the centre of the two circles where the radius of the larger circle is twice the radius of the smaller circle.
 a Prove that $\triangle AOB$ is similar to $\triangle LOM$.
 b What is the ratio $AB : LM$?
- 9 $CDEF$ is a parallelogram. $AD = 3$ cm, $DC = 4$ cm.
 a Prove that $\triangle ADE$ is similar to $\triangle EFB$.
 b What is the ratio $CF : FB$?

